Chapter 6 — Continuous Probability Distributions / Normal Distribution

Chapter 6 is about probabilities for **measurements**.

In Chapter 5, we studied **discrete** random variables, like:

- number of heads
- number of people
- number of successes

In Chapter 6, we study **continuous** random variables, like:

- time
- height
- weight
- temperature
- scores

This chapter mainly focuses on the **normal distribution**, **z-score**, and using the **standard normal table**. The chapter outline includes normal distribution and applications of the normal distribution.

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1. What this chapter is about

Chapter 6 helps you answer questions like:

"What is the probability that a value is less than 900?" "What is the probability that a value is between 1100 and 1225?" "What is the probability that a value exceeds 1500?" "How far is a score from the mean?"

The main tool is:

$$

$$
z = \frac{x - \mu}{\sigma}
$$

$$

This changes a normal value x into a **z-score**.

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2. Continuous Random Variable

Definition

A **continuous random variable** is a variable that can take infinitely many values.

Plain English

It is usually something you **measure**, not count.

It can have decimals.

Tiny examples

Variable	Example value
Time	10.5 minutes
Height	172.3 cm
Weight	68.75 kg
Temperature	37.2°C
Exam score	84.5

How to recognize it in exam questions

Look for words like:

“time” “weight” “height” “length” “temperature” “score” “measurement”

Memory hint

Continuous = **can continue forever between values**.

Example:

Between 1 and 2, you can have:

$$1.1, 1.11, 1.111, 1.1111

and so on.

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3. Normal Distribution

Definition

A **normal distribution** is a continuous probability distribution shaped like a bell.

The chapter says a normally distributed random variable has a relative frequency histogram in the shape of a normal curve, also called a normal density curve.

Plain English

The normal distribution is the famous **bell curve**.

Most values are near the middle. Fewer values are far away from the middle.

Tiny example

Exam scores often look roughly normal:

- Many students score near the average.
- Few students score extremely low.
- Few students score extremely high.

Exam clue

If the question says:

“normally distributed” “normal curve” “mean and standard deviation are…” “find probability less than / greater than / between”

then you are probably using Chapter 6.

Memory hint

Normal distribution = **bell-shaped curve**.

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4. Shape of the Normal Curve

The normal curve looks like this idea:

Low values → middle values → high values

The highest point is in the middle.

The middle is the mean:

$$\mu$$

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5. Mean

$$\mu$$

in Normal Distribution

Definition

The mean

$$\mu$$

$$ is the center of the normal curve.

The chapter says the mean $$

$$\mu$$

is the center of the curve.

Plain English

The mean tells us where the bell curve is centered.

Tiny example

If exam scores are normally distributed with:

$$\mu = 75$$

then the center of the curve is 75.

Exam clue

The question may say:

“mean = 75” “average = 75”

$$\mu = 75$$

All mean the same thing.

Memory hint

Mean = **middle of the bell**.

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6. Standard Deviation

$$\sigma$$

Definition

The standard deviation

$$\sigma$$

tells how spread out the normal curve is.

Plain English

It tells whether the values are close to the mean or far from the mean.

Standard deviation	Meaning
Small σ	values close to mean
Large σ	values spread out

Tiny example

Class A:

$$\mu = 75,\ \sigma = 5$$

$$ Most students are close to 75.

Class B: $$

$$\mu = 75,\ \sigma = 20$$

Scores are much more spread out.

Exam clue

The question may say:

“standard deviation = 10”

$$\sigma = 10$$

Memory hint

Standard deviation = **spread amount**.

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7. Properties of the Normal Distribution

The chapter lists these important properties of the normal density curve: it is symmetric about the mean, mean = median = mode, total area under the curve is 1, and the left and right areas around the mean are equal.

Memorize these:

Property	Simple meaning
Bell-shaped	looks like a bell
Symmetric	left side mirrors right side
Mean = median = mode	center, middle, and highest point are same
Total area = 1	all probabilities together equal 1
Left of mean = right of mean	each side is 0.5
One peak	normal distribution is unimodal

Exam trick

If the question asks:

“Which does **not** apply to normal distribution?”

Then look for something false, like:

bimodal

Bimodal means two peaks. Normal distribution has one peak.

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8. Area Under the Curve

Definition

In a normal distribution, probability is represented by **area under the curve**.

Plain English

To find probability, we find area.

For example:

$P(X < 90)$

means:

Area to the left of 90.

$P(X > 90)$

means:

Area to the right of 90.

$P(80 < X < 90)$

means:

Area between 80 and 90.

Exam clue

Wording	Area
less than	left side
below	left side
greater than	right side
exceeds	right side
more than	right side
between	middle area
at most	left side
at least	right side

Memory hint

Probability = **area**.

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9. Total Area Under the Curve

The total area under the normal curve is:

$$1

That means 100%.

Half is on the left of the mean:

$$0.5

Half is on the right of the mean:

$$0.5

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10. Empirical Rule

The chapter says that for a normal distribution: about 68% of values lie within 1 standard deviation of the mean, 95% within 2 standard deviations, and 99.7% within 3 standard deviations.

Rule

Interval	Approximate percentage

$$

\mu - \sigma

$$

to

$$

\mu + \sigma

$$

| 68% | |

$$

\mu - 2\sigma

$$

to

$$

\mu + 2\sigma

$$

| 95% | |

$$

\mu - 3\sigma

$$

to

$$

\mu + 3\sigma

$$

| 99.7% |

Plain English

Most values are close to the mean.

Very few values are far away.

Tiny example

Suppose:

$$
\mu = 100
$$

$$
\sigma = 10
$$

Then:

68% range

$$100-10=90

$$100+10=110

So about 68% of values are between 90 and 110.

95% range

$$100-2(10)=80

$$100+2(10)=120

So about 95% of values are between 80 and 120.

99.7% range

$$100-3(10)=70

$$100+3(10)=130

So about 99.7% of values are between 70 and 130.

Exam clue

If the question says:

“within one standard deviation” “within two standard deviations” “within three standard deviations”

use the empirical rule.

Memory hint

Memorize:

$$68, 95, 99.7

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11. Standard Normal Distribution

Definition

The **standard normal distribution** is a normal distribution with:

$$\mu = 0$$

$$\sigma = 1$$

The chapter states that the standard normal random variable has mean 0 and standard deviation 1.

Plain English

It is the special normal curve used by the z-table.

Why do we use it?

There are many normal distributions.

Example:

- mean 75, standard deviation 10
- mean 1050, standard deviation 222
- mean 2200, standard deviation 200

Instead of having a separate table for each curve, we convert everything to the standard normal curve using a z-score.

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12. Z-Score

Definition

A **z-score** tells how many standard deviations a value is from the mean.

Formula:

$$z = \frac{x - \mu}{\sigma}$$

The chapter gives this standardization formula for converting X into Z.

Plain English

A z-score answers:

"How far is this value from the average?"

What the sign means

Z-score	Meaning
positive z	value is above the mean
negative z	value is below the mean
z = 0	value equals the mean

Tiny examples

Example 1

$$z = 1.5$$

This means the value is 1.5 standard deviations above the mean.

Example 2

$$z = -0.68$$

This means the value is 0.68 standard deviations below the mean.

Example 3

$$z = 0$$

This means the value is exactly at the mean.

Exam clue

If the question gives:

$$x,\ \mu,\ \sigma$$

$$andasksforz - scoreorprobability, use :$$

$$z = \frac{x - \mu}{\sigma}$$

Memory hint

Z-score = **distance from mean measured in standard deviations**.

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13. How to Use the Z-Table

Most z-tables give:

$P(Z < z)$

This means the area to the **left** of the z-score.

Main rules

Rule 1: Less than

If the question asks:

$P(X < a)$

Convert a to z, then use the table directly.

Rule 2: Greater than

If the question asks:

$P(X > a)$

Convert a to z, then do:

$\$1-$P(Z<z)

Rule 3: Between

If the question asks:

$P(a < X < b)$

Convert both values to z-scores, then subtract:

$P(Z < z_b)$-$\$$P(Z<z_a)

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14. Less Than Problems

Example

Find:

$P(X < 900)$

Suppose:

$\$\$$

$$\mu = 1050, \quad \sigma = 222$$

$\$\$$

Step 1: Identify wording

“Less than” means left side.

Step 2: Convert to z-score

$$
z = \frac{x - \mu}{\sigma}
$$

$$
z = \frac{900 - 1050}{222}
$$

$$
z = \frac{-150}{222}
$$

$$
z = -0.68
$$

Step 3: Use z-table

$P(Z < -0.68)$=0.2483

Final answer

$$0.2483$$

Exam hint

“Less than” means use the left-table value directly.

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15. Greater Than / Exceeds Problems

Example

Find:

$P(X > 1500)$

Suppose:

$$\mu = 1050, \quad \sigma = 222$$

Step 1: Identify wording

“Exceeds” means greater than.

This is right side.

Step 2: Convert to z-score

$$z = \frac{1500 - 1050}{222}$$

$$z = \frac{450}{222}$$

$$z = 2.03$$

Step 3: Use z-table

The z-table gives left side:

$P(Z < 2.03)$=0.9788

Step 4: Right side

$P(Z > 2.03)$=1-0.9788

=0.0212

Final answer

$$0.0212

Exam hint

“Greater than” / “exceeds” means:

$$1-left area

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16. Between Problems

Example

Find:

$P(1100 < X < 1225)$

Suppose:

$$\mu = 1050, \quad \sigma = 222$$

Step 1: Identify wording

“Between” means middle area.

Step 2: Convert lower value to z-score

$$z_1 = \frac{1100 - 1050}{222}$$

$$z_1 = \frac{50}{222}$$

z_1=0.23

Step 3: Convert upper value to z-score

$$z_2 = \frac{1225 - 1050}{222}$$

$$z_2 = \frac{175}{222}$$

z_2=0.79

Step 4: Find left areas from z-table

$P(Z < 0.79)$=0.7852

$P(Z < 0.23)$=0.5909

Step 5: Subtract

$P(0.23 < Z < 0.79)$=0.7852-0.5909

=0.1943

Final answer

$$0.1943

Closest exam option may be:

$$0.1952

Exam hint

“Between” means:

larger left area - smaller left area

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17. Chapter 6 Final Exam Questions

Now we solve the Chapter 6 questions from your final/reference images.

Question 16

Which of the following properties does not apply to a theoretical normal distribution?

Options:

A. The normal distribution is bell-shaped. B. The mean, median, and mode are equal. C. It is a bimodal distribution. D. The area under the normal curve is equal to 1.

Step 1: Identify the topic

This is asking about normal distribution properties.

Step 2: Check option A

Normal distribution is bell-shaped.

This is true.

Step 3: Check option B

For normal distribution:

mean=median=mode

This is true.

Step 4: Check option C

Bimodal means two modes / two peaks.

Normal distribution has one peak.

So this is false.

Step 5: Check option D

Total area under the normal curve is 1.

This is true.

Final answer

C. It is bimodal distribution

Exam clue

Normal distribution is **unimodal**, not bimodal.

Common mistake

Thinking “symmetric” means two peaks. It does not. Symmetric means both sides are mirror images.

Questions 17–19 Given Information

The problem says:

The production time is normally distributed with:

$$\mu = 1050$$

$$\sigma = 222$$

$$We use:$$

$$z = \frac{x - \mu}{\sigma}$$

Question 17

Find the probability that the production time is between 1100 and 1225 minutes.

Options include:

A. 0.7823 B. 0.5871 C. 0.1952 D. None of these answers

Step 1: Identify what is asked

The word **between** means:

$P(1100 < X < 1225)$

This is middle area.

Step 2: Convert 1100 to z-score

Formula:

$$z = \frac{x - \mu}{\sigma}$$

$$Substitute:$$

$$z_1 = \frac{1100 - 1050}{222}$$

$$z_1 = \frac{50}{222}$$

z_1=0.225

$$Round:$$

z_1≈0.23

Step 3: Convert 1225 to z-score

$$z_2 = \frac{1225 - 1050}{222}$$

$$z_2 = \frac{175}{222}$$

z_2=0.788

$$Round:$$

z_2≈0.79

Step 4: Get z-table areas

$P(Z < 0.23)$=0.5909

$P(Z < 0.79)$=0.7852

Step 5: Subtract

For between:

$P(0.23 < Z < 0.79)$=$P(Z < 0.79)$-$P(Z<0.23)

=0.7852-0.5909

=0.1943

Final answer

Closest option:

C. 0.1952

Small differences happen because of rounding.

Exam clue

Between two values = convert both to z, then subtract table values.

Common mistake

Choosing 0.7852.

That is only the area to the left of 1225, not the area between 1100 and 1225.

Question 18

Find the probability that the production time exceeds 1500 minutes.

Options include:

A. 0.9772 B. 0.0228 C. 1.9772 D. None of these answers

Step 1: Identify wording

“Exceeds 1500” means:

$P(X > 1500)$

This is right-tail probability.

Step 2: Convert 1500 to z-score

$$z = \frac{1500 - 1050}{222}$$

$$z = \frac{450}{222}$$

$$z = 2.027$$

$$Round:$$

z≈2.03

Step 3: Use z-table

$P(Z < 2.03)$=0.9788

Depending on rounding/table, it may be close to:

$$0.9772

Step 4: Since we need greater than, subtract from 1

$P(Z > 2.03)$=1-0.9788

$$= 0.0212$$

Closest option:

$$0.0228

Final answer

B. 0.0228

Exam clue

"Exceeds" = greater than = right side.

Right side means:

$$1-table area

Common mistake

Choosing 0.9772.

That is the left side. The question asks for right side.

Question 19

Find the probability that assembly time is less than 900 minutes.

Options include:

A. 0.20 B. 0.45 C. 0.50 D. None of these answers

Step 1: Identify wording

"Less than 900" means:

$P(X < 900)$

This is left-tail probability.

Step 2: Convert 900 to z-score

$$

$$z = \frac{900 - 1050}{222}$$

$$

$$

$$z = \frac{-150}{222}$$

$$

$$

$$z = -0.676$$

$$

Round:

$$z \approx -0.68$$

Step 3: Use z-table

$P(Z < -0.68)$=0.2483

Final answer

$$0.2483

This is not 0.20, 0.45, or 0.50.

So the answer is:

D. None of these answers

Exam clue

“Less than” = left side directly.

Common mistake

Using:

$$1-0.2483

That would give the right side, but the question asks for less than.

Question 20

Scores are normally distributed with mean 75 and standard deviation 10. If a student scored 90, find the z-score.

Options:

A. 1.5 B. 0.5 C. 1 D. None of these answers

Step 1: Identify given values

$$x = 90$$

$$\mu = 75$$

$$\sigma = 10$$

Step 2: Formula

$$z = \frac{x - \mu}{\sigma}$$

Step 3: Substitute

$$z = \frac{90 - 75}{10}$$

Step 4: Calculate

$$z = \frac{15}{10}$$

$$z = 1.5$$

Final answer

A. 1.5

Plain meaning

The student scored 1.5 standard deviations above the mean.

Exam clue

If they only ask for z-score, you do not need a z-table.

Just use:

$$
z = \frac{x-\mu}{\sigma}
$$

Common mistake

Doing:

$$
\frac{75-90}{10} = -1.5
$$

The formula is:

$$
x-\mu
$$

not:

$$
\mu-x
$$

Chapter 6 Mini Cheat Sheet

Topic	Formula / Rule
Z-score	$z = \frac{x-\mu}{\sigma}$
Standard normal mean	$\mu = 0$
Standard normal SD	$\sigma = 1$
Total area under curve	1
Area left of mean	0.5
Area right of mean	0.5
Empirical rule	68%, 95%, 99.7%
Less than	use left area from z-table
Greater than	$1-left area$
Between	upper left area − lower left area
Normal distribution shape	bell-shaped
Mean, median, mode	equal
Normal distribution peak	one peak, unimodal

When You See This Wording, Use This Method

Wording in exam	What to do
“normally distributed”	use z-score
“mean”	μ
“standard deviation”	σ
“find z-score”	$z = \frac{x-\mu}{\sigma}$
“less than”	left area
“below”	left area
“greater than”	right area
“exceeds”	$1-left area$
“between”	subtract two left areas
“within 1 SD”	68%
“within 2 SD”	95%
“within 3 SD”	99.7%
“does not apply to normal distribution”	look for false property

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Most Likely Chapter 6 Exam Question Types

1. Identify properties of normal distribution.
2. Choose which property is false.
3. Calculate z-score.
4. Find probability less than a value.
5. Find probability greater than / exceeds a value.
6. Find probability between two values.
7. Use the empirical rule.
8. Interpret whether a value is above or below average.
9. Recognize that total area under the curve is 1.
10. Understand that mean = median = mode in a normal distribution.

Quick Practice With Answers

Practice 1

A normal distribution has:

$$
\mu = 100, \quad \sigma = 15
$$

Find z-score for:

$$
x = 130
$$

Solution

$$
z = \frac{130 - 100}{15}
$$

$$
z = \frac{30}{15}
$$

$$
z = 2
$$

Answer:

$$2

Meaning: 130 is 2 standard deviations above the mean.

Practice 2

A normal distribution has:

$$
\mu = 50, \quad \sigma = 5
$$

$$
Findz - scorefor:
$$

$$x = 40$$

Solution

$$z = \frac{40 - 50}{5}$$

$$z = \frac{-10}{5}$$

$$z = -2$$

$$Answer:$$

-2

Meaning: 40 is 2 standard deviations below the mean.

Practice 3

If:

$P(Z < 1.20)$=0.8849

$$find:$$

$P(Z > 1.20)$

Solution

$P(Z > 1.20)$=1-0.8849

=0.1151

Answer:

$$0.1151

Practice 4

If:

$P(Z < 1.00)$=0.8413

and:

$P(Z < 0.50)$=0.6915

find:

$P(0.50 < Z < 1.00)$

Solution

$$0.8413-0.6915=0.1498

Answer:

$$0.1498

Practice 5

Which is false about normal distribution?

A. It is bell-shaped. B. Mean = median = mode. C. It has two peaks. D. Total area equals 1.

Answer:

C. It has two peaks.

Normal distribution has one peak.

Final Chapter 6 Memory Box

Memorize this:

$$
z = \frac{x - \mu}{\sigma}
$$

Then remember:

Less than = left area Greater than = 1 − left area Between = subtract two left areas

Also memorize normal properties:

bell-shaped, symmetric, mean = median = mode, total area = 1, one peak.